THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · HARD LEVEL CHAPTER TEST SET 1

Chapter 8: **Sound Waves: Characteristics and Applications** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Assertion (A): Sound cannot travel in a vacuum. Reason (R): Sound waves need a material medium for propagation. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

2. A sound wave travels from air (340 m/s) into water (1500 m/s). A 500 Hz wave has wavelengths in air and water of: 1

(a) 3 m and 0.68 m **(b)** 0.68 m and 0.68 m
(c) 3 m and 3 m **(d)** 0.68 m and 3 m

3. If the temperature rises from 0 degrees Celsius to 40 degrees Celsius, the speed of sound in air increases from 331 m/s by about (0.6 m/s per degree Celsius): 1

(a) 40 m/s **(b)** 240 m/s **(c)** 24 m/s **(d)** 6 m/s

4. A sonar signal gives echoes from two surfaces at 0.8 s and 1.0 s. The separation between the surfaces (v = 1500 m/s) is: 1

(a) 600 m **(b)** 75 m **(c)** 300 m **(d)** 150 m

5. Assertion (A): The speed of sound is greater in solids than in gases. Reason (R): The particles in solids are closely packed and strongly bound. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. A person standing at 340 m from a cliff fires a gun. The echo is heard after: 1

(a) 1 s **(b)** 0.5 s **(c)** 4 s **(d)** 2 s

7. A sound of 20 degrees Celsius air moves at 344 m/s. The distance travelled in 0.25 s is: 1

(a) 86 m **(b)** 43 m **(c)** 1376 m **(d)** 172 m

8. A pulse takes 0.1 s to travel along a 34 m long steel pipe. The speed in steel is: 1

(a) 340 m/s **(b)** 34 m/s **(c)** 0.0029 m/s **(d)** 3400 m/s

9. **Assertion (A):** Sound cannot travel through a vacuum. 1
Reason (R): Sound is a mechanical wave and needs a material medium for its propagation.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** Sound waves in air are longitudinal waves. 1
Reason (R): The speed of sound is greater in solids than in gases.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A boy standing 85 m in front of a hill claps and hears the echo after 0.5 s. Find the speed of sound. 2

12. Why are two sounds of the same loudness and pitch from a violin and a sitar distinguishable? 2

13. State why the sound of an approaching train is first heard through the rails. 2

14. The speed of sound in steel is 5000 m/s. A worker strikes one end of a 1 km steel rail. Find the time interval between the sounds heard through steel and air (v = 340 m/s) at the other end. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Two sound waves have frequencies 100 Hz and 400 Hz with equal amplitude. Compare their wavelengths, speeds, loudness and pitch in the same medium. 3

16. A siren emits a sound of frequency 700 Hz. Find the time period, the wavelength in air (340 m/s), and the number of waves reaching a man in 5 s. 3

17. Explain why the sound of a bell is heard for a long time in a stone temple but only for a short time in an open field. 3

OR

A man fires a gun in front of a cliff and hears an echo after 2 s and a second after 3 s from another cliff. Find the distance between the cliffs if he stands between them (v = 340 m/s). 3

18. A man claps near a big cliff and hears the echo after 3 s. He then walks 85 m toward the cliff and claps again. Find the new echo time (v = 340 m/s). 3

OR

Describe the working of an electric bell in a vacuum jar and a microphone as a transducer. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A research ship in the Arabian Sea sends an ultrasonic pulse downward from its SONAR and receives the echo from the sea bed after 0.8 s. The speed of sound in sea water is 1500 m/s. 5
(i) Expand the term SONAR.
(ii) Find the total distance travelled by the pulse.
(iii) Find the depth of the sea.
(iv) Find the depth at another spot where the echo returns after 1.2 s, and state one reason for using ultrasound in SONAR.

20. Describe an experiment to measure the speed of sound in air using two observers and a gun or a clapper at a large distance. State the observations, the calculation for 1020 m with a time of 3 s, and the precautions. 5

OR

Describe how ultrasound is used in the SONAR, in medical sonography and in non-destructive testing, including any limitations. 5

END OF QUESTION PAPER — Total: 40 Marks